

# Traffic Vehicle Classification

Model evaluation report - validation and test performance of all 13 trained models

| field                        | value                                                                 |
|------------------------------|-----------------------------------------------------------------------|
| generated                    | 2026-10-08 18:50:01                                                   |
| config                       | configs/default.json                                                  |
| run fingerprint              | 1ba0f75a8c63be033d34e0cc95f1a600081d2d887b9763e64850f8adddd401        |
| classes                      | 8: ambulance, autobus, kamyun, kamyunet, minibus, savari, taxi, vanet |
| dataset split                | 930 train / 232 validation / 64 test                                  |
| models                       | 13 experiments, all 200 epochs                                        |
| production model             | ResNet18 fine-tuning (resnet18_fine_tuning)                           |
| production checkpoint sha256 | 8e1147b5ee5108737ac3a1ad92704e7af3e08215b6d6f81b78f3e55545a7c6d3      |
| selection protocol           | best validation macro-F1; test split used for reporting only          |

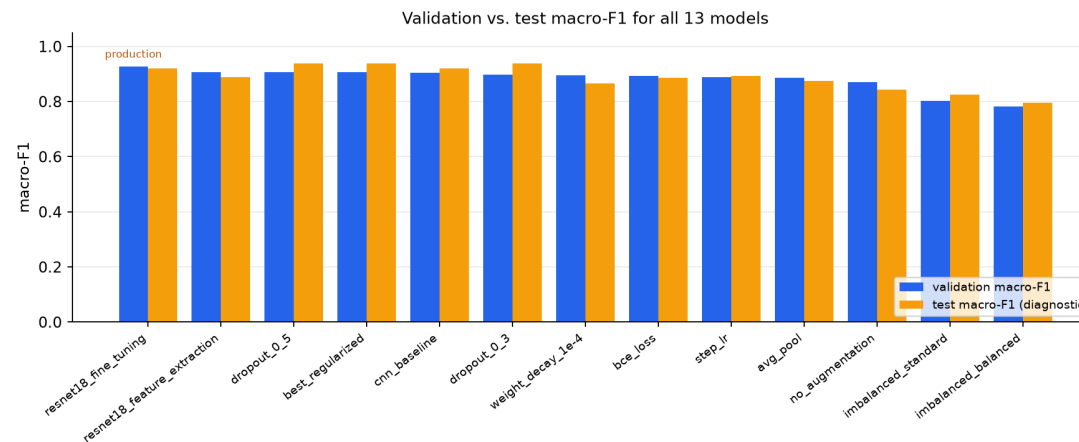
## Key findings

- Highest validation macro-F1: **ResNet18 fine-tuning** (0.9274 at epoch 125).
- Selected production model: **ResNet18 fine-tuning** (validation macro-F1 0.9274, test macro-F1 0.9209 on 64 images, 5 errors).
- Highest test macro-F1 (diagnostic): Dropout p=0.3, Dropout p=0.5, Best regularized + scheduled (0.9381).
- Validation macro-F1 spans 0.7825 to 0.9274 across 13 models (spread 0.1449).
- Production test performance: accuracy 0.9219, macro-F1 0.9209, 5 errors.
- Selection protocol: validation macro-F1 only -> resnet18\_fine\_tuning; 1 model(s) evaluated on the test split.

# 1. Results overview (validation and test)

Sorted by validation macro-F1. Test columns come from the post-selection diagnostic over the 64-image test split.

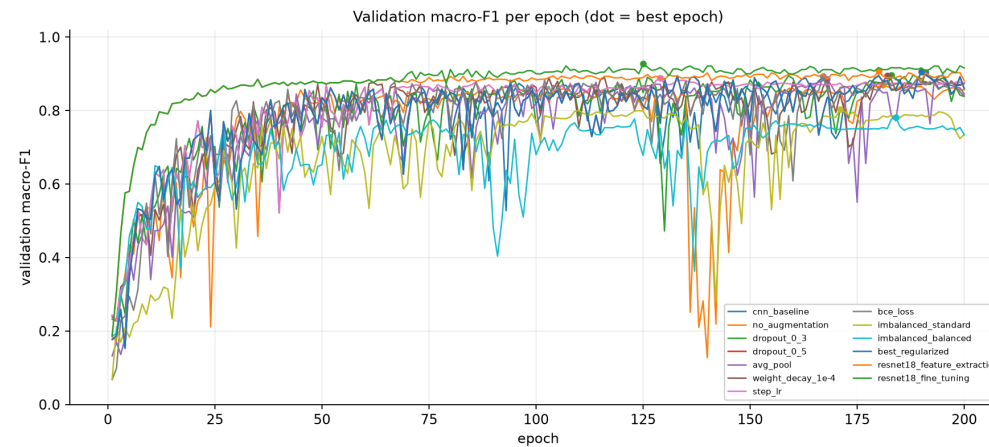
| model                         | category        | arch            | epoch      | val acc       | val macro-F1  | test acc      | test macro-F1 | test errors | test mean conf. |
|-------------------------------|-----------------|-----------------|------------|---------------|---------------|---------------|---------------|-------------|-----------------|
| <b>ResNet18 fine-tuning</b>   | <b>transfer</b> | <b>resnet18</b> | <b>125</b> | <b>0.9267</b> | <b>0.9274</b> | <b>0.9219</b> | <b>0.9209</b> | <b>5/64</b> | <b>0.8892</b>   |
| ResNet18 feature extraction   | transfer        | resnet18        | 180        | 0.9095        | 0.9072        | 0.8906        | 0.8878        | 7/64        | 0.8602          |
| Dropout p=0.5                 | ablation        | cnn             | 190        | 0.9052        | 0.9059        | 0.9375        | 0.9381        | 4/64        | 0.9569          |
| Best regularized + scheduled  | main            | cnn             | 190        | 0.9052        | 0.9059        | 0.9375        | 0.9381        | 4/64        | 0.9569          |
| CNN baseline                  | ablation        | cnn             | 191        | 0.9052        | 0.9047        | 0.9219        | 0.9213        | 5/64        | 0.9430          |
| Dropout p=0.3                 | ablation        | cnn             | 183        | 0.8966        | 0.8966        | 0.9375        | 0.9381        | 4/64        | 0.9529          |
| Weight decay 1e-4             | ablation        | cnn             | 182        | 0.8966        | 0.8956        | 0.8594        | 0.8652        | 9/64        | 0.9446          |
| BCEWithLogitsLoss             | ablation        | cnn             | 167        | 0.8922        | 0.8930        | 0.8906        | 0.8863        | 7/64        | 0.8797          |
| StepLR                        | ablation        | cnn             | 129        | 0.8879        | 0.8888        | 0.8906        | 0.8931        | 7/64        | 0.9422          |
| Average pooling               | ablation        | cnn             | 168        | 0.8836        | 0.8859        | 0.8750        | 0.8752        | 8/64        | 0.9379          |
| No augmentation               | ablation        | cnn             | 181        | 0.8707        | 0.8702        | 0.8438        | 0.8435        | 10/64       | 0.9098          |
| Imbalanced + standard batches | imbalance       | cnn             | 171        | 0.8017        | 0.8032        | 0.8281        | 0.8262        | 11/64       | 0.8779          |
| Imbalanced + balanced batches | imbalance       | cnn             | 184        | 0.7845        | 0.7825        | 0.7969        | 0.7962        | 13/64       | 0.8889          |



## 2. Validation evaluation (232 images, all models)

Metrics at each model's best epoch, selected on the validation macro-F1. ECE is the 10-bin expected calibration error of the raw (uncalibrated) validation probabilities.

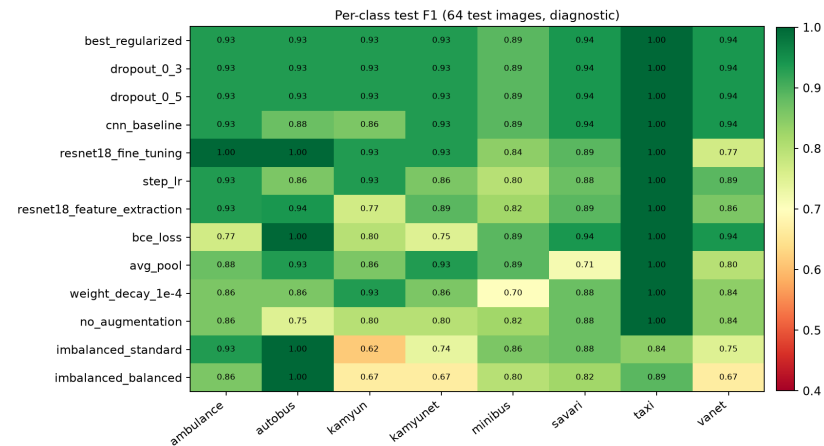
| model                         | loss                 | epoch      | acc           | macro-P       | macro-R       | macro-F1      | weighted-F1   | ECE           | mean conf.    | elapsed s   |
|-------------------------------|----------------------|------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|-------------|
| <b>ResNet18 fine-tuning</b>   | <b>cross_entropy</b> | <b>125</b> | <b>0.9267</b> | <b>0.9294</b> | <b>0.9287</b> | <b>0.9274</b> | <b>0.9265</b> | <b>0.0636</b> | <b>0.9627</b> | <b>2279</b> |
| ResNet18 feature extraction   | cross_entropy        | 180        | 0.9095        | 0.9112        | 0.9081        | 0.9072        | 0.9082        | 0.0788        | 0.8726        | 2216        |
| Dropout p=0.5                 | cross_entropy        | 190        | 0.9052        | 0.9137        | 0.9033        | 0.9059        | 0.9056        | 0.0583        | 0.9551        | 2534        |
| Best regularized + scheduled  | cross_entropy        | 190        | 0.9052        | 0.9137        | 0.9033        | 0.9059        | 0.9056        | 0.0583        | 0.9551        | 2664        |
| CNN baseline                  | cross_entropy        | 191        | 0.9052        | 0.9094        | 0.9043        | 0.9047        | 0.9046        | 0.0813        | 0.9471        | 2524        |
| Dropout p=0.3                 | cross_entropy        | 183        | 0.8966        | 0.9076        | 0.8950        | 0.8966        | 0.8966        | 0.0845        | 0.9540        | 2587        |
| Weight decay 1e-4             | cross_entropy        | 182        | 0.8966        | 0.9018        | 0.8946        | 0.8956        | 0.8957        | 0.0535        | 0.9494        | 2737        |
| BCEWithLogitsLoss             | bce                  | 167        | 0.8922        | 0.8947        | 0.8922        | 0.8930        | 0.8923        | 0.1122        | 0.8840        | 3131        |
| StepLR                        | cross_entropy        | 129        | 0.8879        | 0.8960        | 0.8848        | 0.8888        | 0.8888        | 0.0740        | 0.9431        | 2678        |
| Average pooling               | cross_entropy        | 168        | 0.8836        | 0.8959        | 0.8819        | 0.8859        | 0.8839        | 0.0812        | 0.9459        | 2570        |
| No augmentation               | cross_entropy        | 181        | 0.8707        | 0.8726        | 0.8705        | 0.8702        | 0.8703        | 0.0829        | 0.9242        | 2546        |
| Imbalanced + standard batches | cross_entropy        | 171        | 0.8017        | 0.8373        | 0.8046        | 0.8032        | 0.8010        | 0.1215        | 0.9050        | 2706        |
| Imbalanced + balanced batches | cross_entropy        | 184        | 0.7845        | 0.8224        | 0.7848        | 0.7825        | 0.7839        | 0.1559        | 0.9218        | 2397        |



### 3. Test evaluation (64 images, diagnostic)

The test split was never used for model selection or tuning; these numbers are a post-selection diagnostic. Ranked by test macro-F1.

| model                         | acc           | macro-P       | macro-R       | macro-F1      | errors      | mean conf.    | below review thr. | top confusion pair          | official   |
|-------------------------------|---------------|---------------|---------------|---------------|-------------|---------------|-------------------|-----------------------------|------------|
| Dropout p=0.3                 | 0.9375        | 0.9472        | 0.9375        | 0.9381        | 4/64        | 0.9529        | 7                 | ambulance/vanet (0.125)     | no         |
| Dropout p=0.5                 | 0.9375        | 0.9472        | 0.9375        | 0.9381        | 4/64        | 0.9569        | 6                 | ambulance/vanet (0.125)     | no         |
| Best regularized + scheduled  | 0.9375        | 0.9472        | 0.9375        | 0.9381        | 4/64        | 0.9569        | 6                 | ambulance/vanet (0.125)     | no         |
| CNN baseline                  | 0.9219        | 0.9316        | 0.9219        | 0.9213        | 5/64        | 0.9430        | 6                 | ambulance/vanet (0.125)     | no         |
| <b>ResNet18 fine-tuning</b>   | <b>0.9219</b> | <b>0.9409</b> | <b>0.9219</b> | <b>0.9209</b> | <b>5/64</b> | <b>0.8892</b> | <b>17</b>         | <b>savari/vanet (0.250)</b> | <b>yes</b> |
| StepLR                        | 0.8906        | 0.9177        | 0.8906        | 0.8931        | 7/64        | 0.9422        | 8                 | autobus/minibus (0.250)     | no         |
| ResNet18 feature extraction   | 0.8906        | 0.9083        | 0.8906        | 0.8878        | 7/64        | 0.8602        | 18                | kamyun/kamyunet (0.250)     | no         |
| BCEWithLogitsLoss             | 0.8906        | 0.8981        | 0.8906        | 0.8863        | 7/64        | 0.8797        | 12                | ambulance/kamyun (0.125)    | no         |
| Average pooling               | 0.8750        | 0.8969        | 0.8750        | 0.8752        | 8/64        | 0.9379        | 10                | savari/vanet (0.375)        | no         |
| Weight decay 1e-4             | 0.8594        | 0.8982        | 0.8594        | 0.8652        | 9/64        | 0.9446        | 6                 | autobus/minibus (0.250)     | no         |
| No augmentation               | 0.8438        | 0.8555        | 0.8438        | 0.8435        | 10/64       | 0.9098        | 13                | kamyun/minibus (0.250)      | no         |
| Imbalanced + standard batches | 0.8281        | 0.8486        | 0.8281        | 0.8262        | 11/64       | 0.8779        | 18                | kamyun/kamyunet (0.250)     | no         |
| Imbalanced + balanced batches | 0.7969        | 0.8360        | 0.7969        | 0.7962        | 13/64       | 0.8889        | 16                | kamyun/kamyunet (0.375)     | no         |



#### 4. Production model in detail: ResNet18 fine-tuning

| class     | val support | val P  | val R  | val F1 | test support | test P | test R | test F1 |
|-----------|-------------|--------|--------|--------|--------------|--------|--------|---------|
| ambulance | 25          | 1.0000 | 1.0000 | 1.0000 | 8            | 1.0000 | 1.0000 | 1.0000  |
| autobus   | 28          | 0.9655 | 1.0000 | 0.9825 | 8            | 1.0000 | 1.0000 | 1.0000  |
| kamyun    | 27          | 0.9130 | 0.7778 | 0.8400 | 8            | 1.0000 | 0.8750 | 0.9333  |
| kamyunet  | 29          | 0.7941 | 0.9310 | 0.8571 | 8            | 1.0000 | 0.8750 | 0.9333  |
| minibus   | 28          | 0.9655 | 1.0000 | 0.9825 | 8            | 0.7273 | 1.0000 | 0.8421  |
| savari    | 30          | 0.9000 | 0.9000 | 0.9000 | 8            | 0.8000 | 1.0000 | 0.8889  |
| taxi      | 27          | 0.9259 | 0.9259 | 0.9259 | 8            | 1.0000 | 1.0000 | 1.0000  |
| vanet     | 38          | 0.9714 | 0.8947 | 0.9315 | 8            | 1.0000 | 0.6250 | 0.7692  |

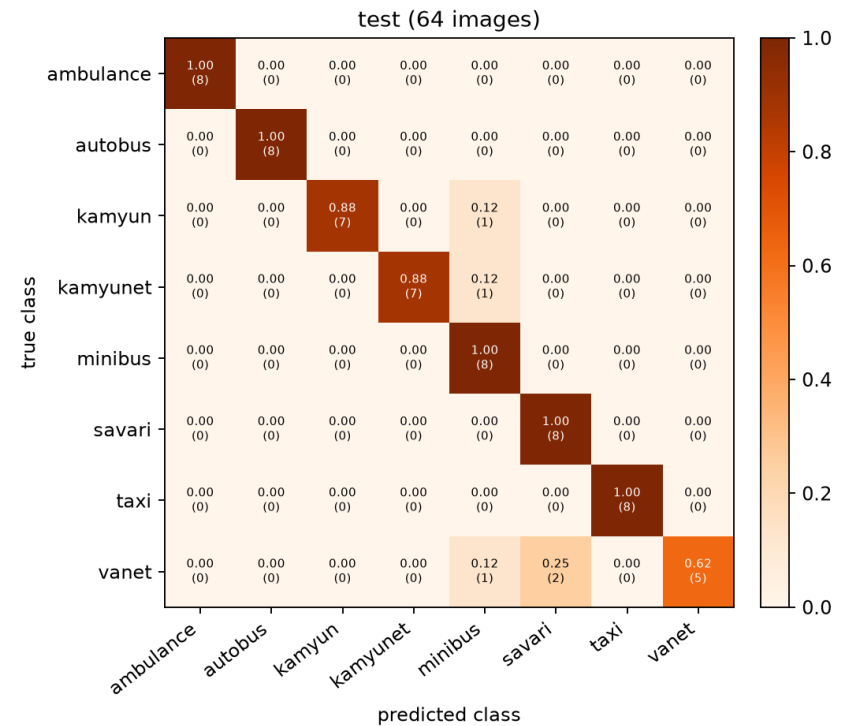
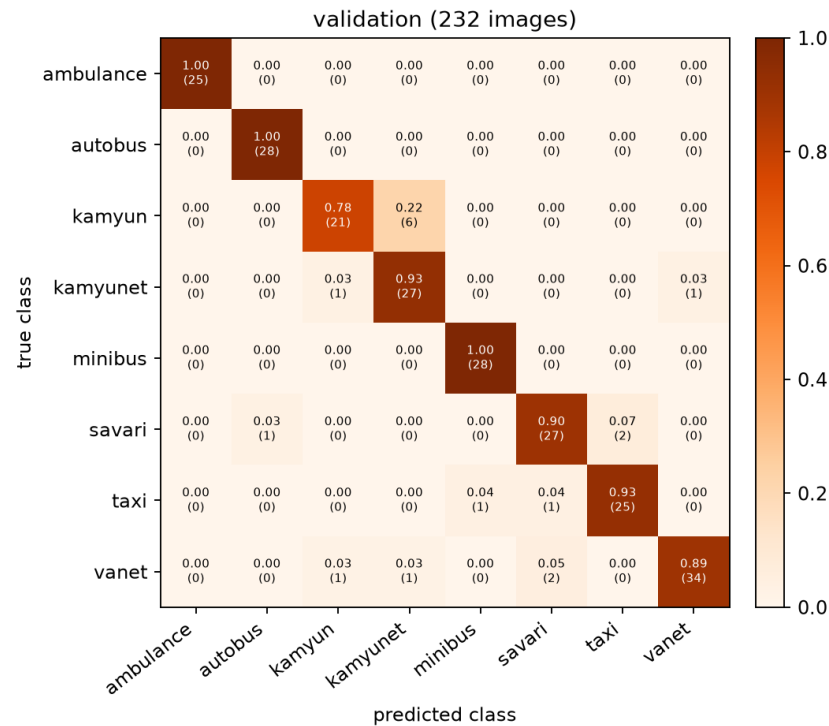
#### Calibration and uncertainty

| split / state                  | NLL    | ECE (10 bins) | Brier  | mean confidence | accuracy |
|--------------------------------|--------|---------------|--------|-----------------|----------|
| validation, raw                | 0.4426 | 0.0636        | 0.1398 | 0.9627          | 0.9267   |
| temperature = 2.115            | -      | -             | -      | -               | -        |
| validation, temperature-scaled | 0.3105 | 0.0438        | 0.1410 | 0.9004          | 0.9267   |
| test, temperature-scaled       | 0.3109 | 0.0700        | 0.1557 | 0.8892          | 0.9219   |

## Selective prediction on the test split

| field                                     | value   |
|-------------------------------------------|---------|
| review threshold (selected on validation) | 0.8393  |
| accepted / review                         | 47 / 17 |
| automatic coverage                        | 0.7344  |
| review rate                               | 0.2656  |
| automatic accuracy                        | 0.9787  |
| review accuracy                           | 0.7647  |
| errors flagged for review                 | 4 / 5   |

## 5. Confusion matrices - production model



### Most frequent mutual confusions (test split)

| class pair            | A as B | B as A | pair confusion |
|-----------------------|--------|--------|----------------|
| savari <-> vanet      | 0.0    | 0.25   | 0.2500         |
| kamyun <-> minibus    | 0.125  | 0.0    | 0.1250         |
| kamyunet <-> minibus  | 0.125  | 0.0    | 0.1250         |
| minibus <-> vanet     | 0.0    | 0.125  | 0.1250         |
| ambulance <-> autobus | 0.0    | 0.0    | 0.0000         |

## 6. Methodology and caveats

- Model selection used the validation split only (stratified 20% of the training images, seed 42); the best epoch of each run is chosen by validation macro-F1.
- The test split is evaluated after selection, once per model, and is reported here as a diagnostic. It must not be used to pick a model or tune hyperparameters.
- All models share the same class mapping, stratified split and evaluation transform; differences between rows are attributable to the training configuration named by each experiment.
- ECE uses 10 equal-width confidence bins; production-model NLL/Brier are computed after temperature scaling, with the temperature fitted on the validation split.
- Integrity check: recomputed test macro-F1 matches the locked value (abs difference 0.0).
- Per-image test predictions for non-production models are diagnostic outputs of scripts/evaluate\_all\_models\_on\_test.py; their per-class F1 values are recomputed from those predictions.

Generated by scripts/generate\_pdf\_report.py on 2026-10-08 18:50:03.